Spatio-Temporal Cluster Detection of Forest Fires in Southern Peru (2017–2025) Using the SaTScan Method and Spatial Autocorrelation

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Abstract

Forest fires are a growing threat to Peru's Andean-Amazonian ecosystems. This study detects and characterizes spatio-temporal clusters of forest fires in southern Peru (Madre de Dios, Puno, and Cusco) during 2017–2025, integrating 74,861 satellite-derived thermal anomalies (NASA FIRMS VIIRS/MODIS) and 2,727,647 deforestation alerts (GeoBosques-MINAM) at the district and monthly level. A discrete Poisson spatio-temporal scan model (SaTScan) was applied, complemented by global (Moran's Index) and local (univariate and bivariate LISA) spatial autocorrelation analysis, together with sensitivity tests using 25 %, 30 %, and 50 % spatial windows. Results reveal extreme seasonality, with 89.51 % of events concentrated between July and October (peak in September: 39.01 %), and two interannual maxima coinciding with severe droughts and El Niño episodes (2020 and 2024). Nine statistically significant spatio-temporal clusters were identified ($p < 0.00001$). The primary cluster was located at the Cusco–Puno Andean interface (96 districts, 2019–2022, LLR = 35,231.2, RR = 780.02), whereas Madre de Dios concentrated the highest absolute fire burden within Amazonian clusters (Cluster 4: 14,264 fires; Cluster 6: 15,863 fires). The bivariate Moran's Index ($I = 0.6019$, $p = 0.001$) confirmed a strong spatial dependence between prior deforestation and fire occurrence in neighboring districts, and sensitivity analyses confirmed the full stability of the primary cluster. These findings provide key empirical evidence for prioritizing intervention zones and early-warning systems by Peru's forestry and environmental authorities (SERFOR and MINAM).

Keywords: Spatial statistics, SaTScan, Forest fires, Deforestation, Spatial autocorrelation, LISA, Southern Peru

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1. Introduction

Forest fires constitute one of the main environmental threats to the Amazonian and Andean ecosystems of South America, with direct consequences for biodiversity, air quality, the health of local populations, and greenhouse gas emissions (Aragão et al., 2018; Nepstad et al., 1999; Cochrane, 2003). In Peru, the frequency and intensity of these events have intensified over the last decade, driven by the expansion of the agricultural frontier, traditional burning practices for land-use change, and prolonged drought periods associated with climatic phenomena such as El Niño (Servicio Nacional Forestal y de Fauna Silvestre (SERFOR), 2023; Móstiga et al., 2026; Marengo et al., 2011). The southern regions of the country —Madre de Dios, Puno, and Cusco— concentrate a significant share of these events owing to their vast Amazonian forest cover and the persistence of agropastoral burning practices in forest-human interface zones (Ministerio del Ambiente (MINAM), 2024; Barboza et al., 2024; Zubieta et al., 2021). At the continental level, the 2019 fire crisis revealed severe positive anomalies in the Peruvian Amazon and the central Andes, with burned area exceeding the historical average by 70 % (Lizundia-Loiola et al., 2020). Recent studies in the Brazilian Amazon confirm that proximity to roads and pasture expansion are the main anthropogenic drivers of spatio-temporal fire clusters (Valente and Laurini, 2023; Raiol et al., 2026; Bolaño-Díaz et al., 2022).

Spatial statistics has proven to be an effective tool for characterizing the geographic distribution of environmental phenomena and detecting non-random clustering patterns in space and time. Among the available methods, SaTScan —based on spatial and spatio-temporal scanning using discrete probability models— allows the identification of statistically significant clusters of events as well as their temporal evolution, which is particularly useful for anticipating the expansion of fire foci and prioritizing intervention areas (Kulldorff, 1997, 2001; Vega Orozco et al., 2012; Urfalı and Eymen, 2025; Zerbe et al., 2022). This method has been successfully applied to forest fires in Europe and the Brazilian Amazon (Vega Orozco et al., 2012; Valente and Laurini, 2023), although its use in the Peruvian context —outside the domain of public health— remains scarce. In addition, Moran's Index and Local Indicators of Spatial Association (LISA) allow the identification of hot spots and significant spatial anomalies in environmental phenomena such as forest fires, including in bivariate form to relate two contiguous geospatial variables (Anselin, 1995; Ma et al., 2022; Atalay et al., 2025).

In this context, the present study aims to detect and characterize the spatio-temporal clusters of forest fires that occurred in southern Peru during the period 2017–2025, by applying the SaTScan method and spatial autocorrelation indicators (Moran and LISA), contrasting their location with drought periods and forest-cover-loss alerts, in order to provide useful evidence for prioritizing fire prevention and response strategies in the region.

2. Study Area and Data Sources

2.1. Study Area

The study area covers the southern sector of Peru, comprising the administrative regions of Madre de Dios, Puno, and Cusco (Figure 1). This macro-region transitions from the hyperdiverse Amazonian lowlands (Madre de Dios) to the Andean mountain range and the Lake Titicaca basin (Cusco and Puno), encompassing a political-administrative division of **237 districts**.

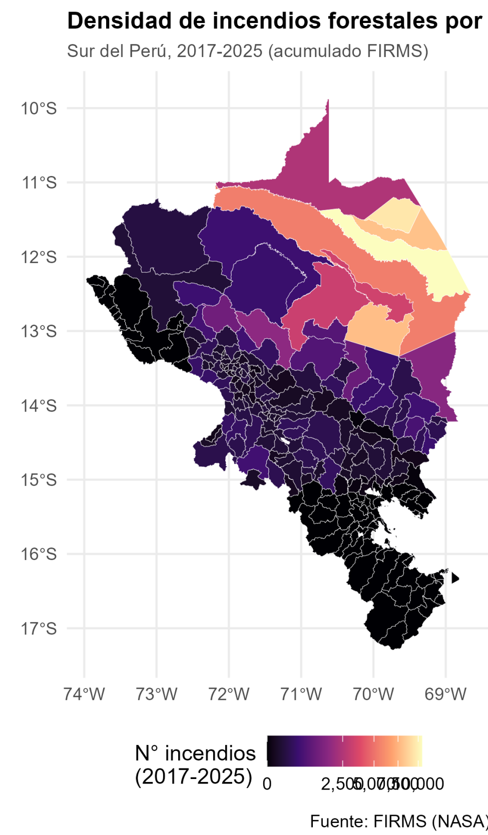

Figure 1: Absolute density map of forest fires by district in southern Peru (2017–2025).

2.2. Fire Hotspots (FIRMS)

Georeferenced point records from the VIIRS (SNPP/NOAA-20) and MODIS sensors were obtained from NASA's FIRMS system for the period from January 1, 2017 to December 31, 2025 (Giglio et al., 2016, 2018; Schroeder et al., 2014). Each hotspot record contains geographic coordinates, acquisition date (*acq_date*), brightness, and confidence level. After spatial aggregation and filtering within the district boundaries of the study area to isolate genuine forest-burning events (Deng et al., 2025; Su et al., 2023), a total of **74,861 fire hotspots** were consolidated.

2.3. Deforestation Alerts (GeoBosques)

To represent the population at risk and land-use-change pressure, spatio-temporal data from MINAM's GeoBosques platform were processed. A total of **2,727,647 point deforestation alerts** occurring between 2017 and 2025 were consolidated. The distribution by department shows that Madre de Dios concentrates 71.77 % of the total (1,957,718 alerts), followed by Cusco (18.31 %, 499,521 alerts) and Puno (9.91 %, 270,408 alerts) (Alarcon-Aguirre et al., 2022).

3. Spatial-Statistical Methodology

3.1. SaTScan Discrete Poisson Model

Spatio-temporal cluster detection was carried out using SaTScan v10.3.3 with **Kulldorff's discrete Poisson spatio-temporal model** (Kulldorff, 1997, 2001; Aboushady et al., 2025). This model has been validated for detecting spatio-temporal clusters of forest fires from satellite hotspot data, allowing structural (recurrent) fire regimes to be distinguished from extemporaneous episodes (Vega Orozco et al., 2012; Levin-Rector et al., 2024). The number of fires c_{it} in district i and month t is assumed to be Poisson distributed with expected parameter:

$$\mu_{it} = p_{it} \cdot \frac{C}{P},$$

where p_{it} is the number of deforestation alerts (baseline risk), C is the total number of fires across the whole study, and P is the total accumulated deforestation.

The spatio-temporal cylindrical scanning window Z varies continuously in space (centroid and radius of the base circle) and in time (range of months). For each cylinder Z, the likelihood function under the alternative hypothesis of increased risk is:

$$L(Z) = \left(\frac{c_Z}{\mu_Z}\right)^{c_Z} \left(\frac{C - c_Z}{C - \mu_Z}\right)^{C - c_Z} \mathbf{1}(c_Z > \mu_Z), \tag{1}$$

where c_Z is the number of observed fires within Z, μ_Z is the expected number, and $\mathbf{1}(\cdot)$ is the indicator function for excess risk. The Log-Likelihood Ratio statistic (LLR) is calculated as:

$$\text{LLR} = \ln\left(\frac{L(Z)}{L_0}\right) = c_Z \ln\left(\frac{c_Z}{\mu_Z}\right) + (C - c_Z) \ln\left(\frac{C - c_Z}{C - \mu_Z}\right). \tag{2}$$

Statistical significance was determined using 999 Monte Carlo simulations. The Relative Risk (RR) was estimated as:

$$\mathrm{RR} = \frac{c_Z/\mu_Z}{(C - c_Z)/(C - \mu_Z)}. \tag{3}$$

3.2. Global and Local Spatial Autocorrelation (Moran and LISA)

In addition, static spatial dependence was assessed using the univariate and bivariate **Global Moran's Index** (I) (Anselin, 1995):

$$I = \frac{N}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_{i=1}^{N} \sum_{j=1}^{N} w_{ij}(x_i - \bar{x})(y_j - \bar{y})}{\sum_{i=1}^{N} (x_i - \bar{x})^2}, \tag{4}$$

where w_{ij} represents the row-standardized spatial contiguity matrix (Queen criterion). To identify the exact location of hot spots (*High-High*) and spatial anomalies, univariate and bivariate Local Indicators of Spatial Association (LISA) were calculated with 999 random permutations (Kumari et al., 2025; Mupfiga et al., 2022; Memisoglu Baykal, 2025). Spatial processing and modeling were carried out in the R statistical environment (R Core Team, 2024), using specialized packages for spatial dependence analysis (Bivand et al., 2013; Bivand and Wong, 2018).

3.3. Sensitivity Analysis

To evaluate the robustness of the detected clusters under different maximum spatial extent assumptions, the SaTScan analysis was repeated restricting the maximum spatial window to 25 % and 30 % of the population at risk, complementing the baseline 50 % configuration.

4. Results

4.1. Temporal Patterns and Seasonality

The temporal analysis (Figure 2) shows marked interannual variability, with two severe critical peaks in **2020** (12,811 fires) and **2024** (12,437 fires), coinciding with periods of extreme drought associated with El Niño events. The aggregated monthly distribution (Figures 3 and 4) reveals extreme concentration during the austral dry season: the months of July through October account for **89.51 %** of annual occurrence, peaking in **September (39.01 %)** and **August (25.22 %)**.

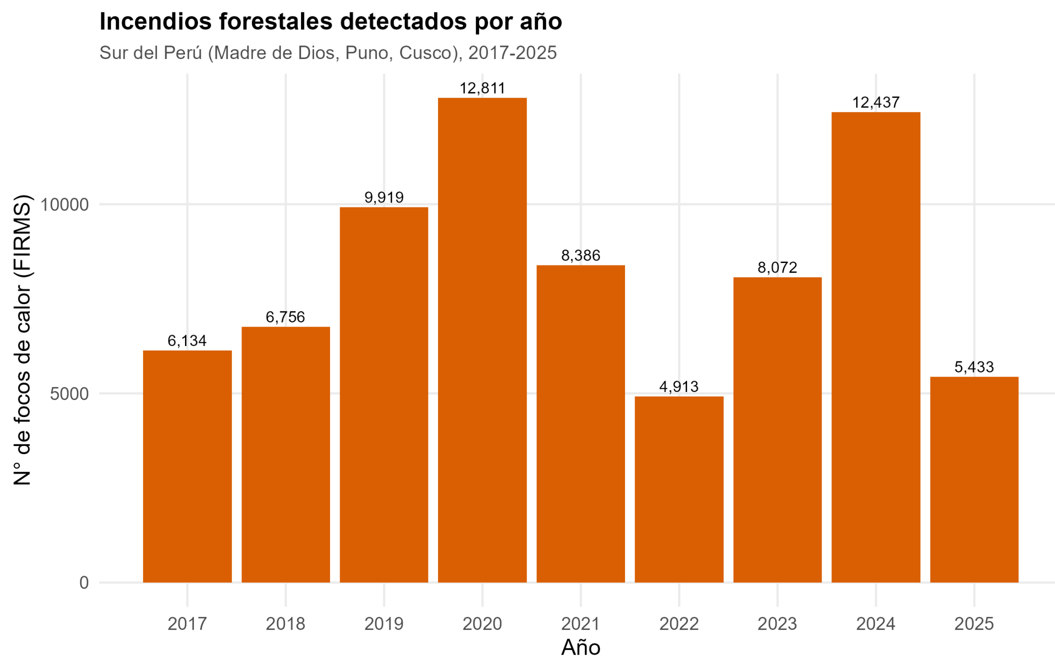

Figure 2: Annual trend of forest fires in southern Peru (2017–2025).

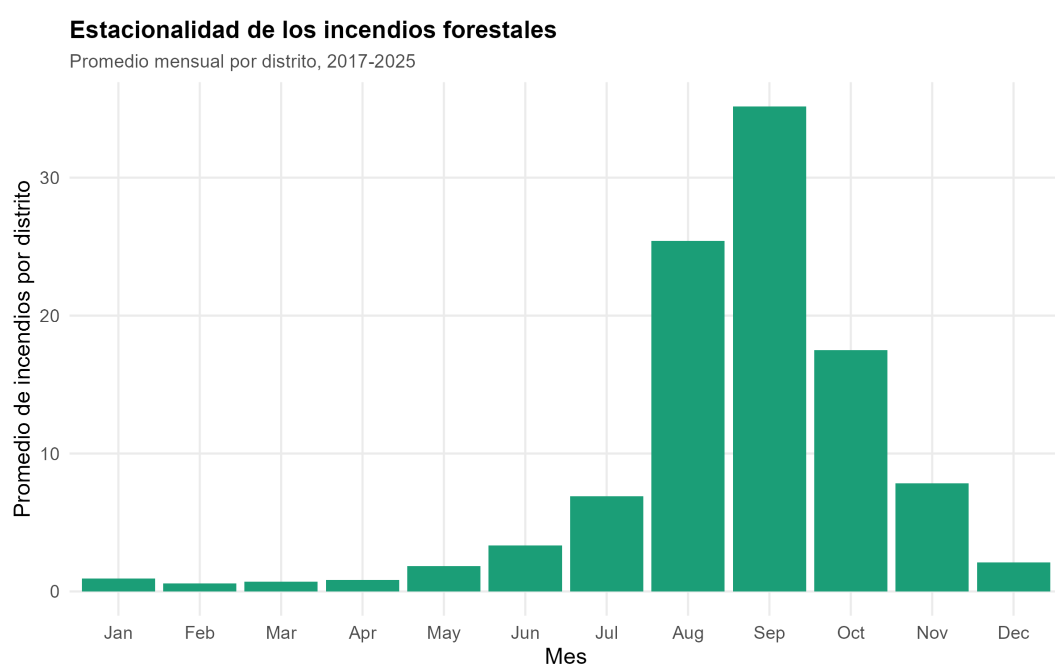

Figure 3: Aggregated monthly distribution and seasonal percentage of forest fires.

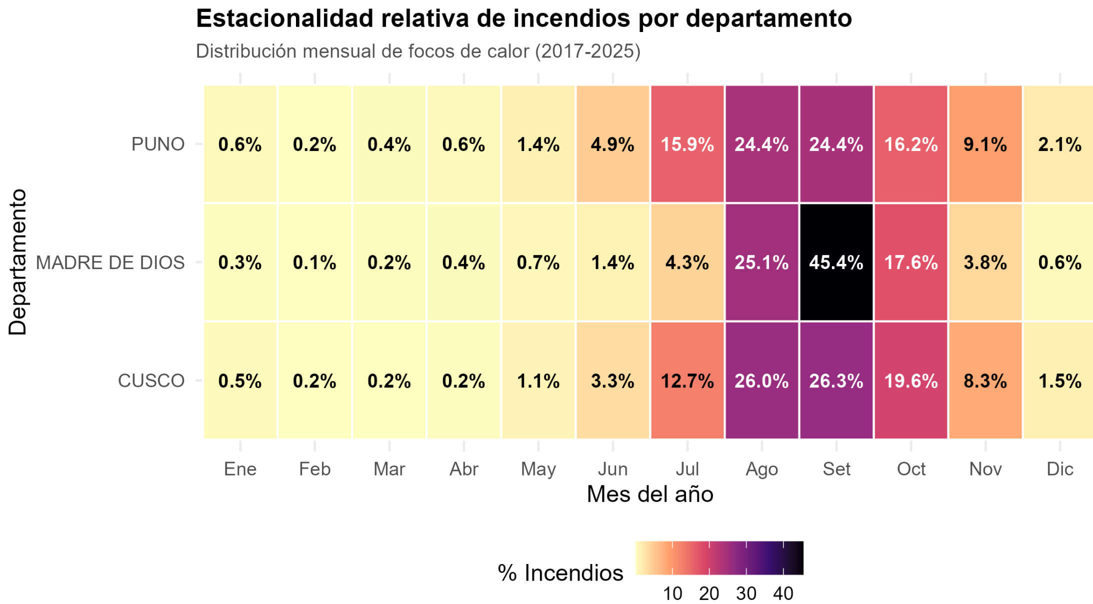

Figure 4: Comparative heat map of monthly seasonality by department (Madre de Dios, Cusco, and Puno).

4.2. Global Spatial Autocorrelation and LISA Maps

The Global Moran's test showed highly significant positive spatial autocorrelation for both forest fires ($I = 0.6407$, $z = 16.918$, $p < 0.00001$) and deforestation ($I = 0.5905$, $z = 15.985$, $p < 0.00001$). The **bivariate Moran's Index** between fires and prior deforestation reached $I = 0.6019$ ($p = 0.001$), demonstrating that forest loss in neighboring districts directly increases fire risk in the focal district (Figure 5).

The univariate (Figure 6) and bivariate (Figure 7) LISA maps confirm the existence of statistically significant *High-High* zones in northern Madre de Dios and at the Cusco–Puno interface, corresponding to the districts with the highest fire concentration and the strongest local association with surrounding deforestation.

4.3. Spatio-Temporal Cluster Detection (SaTScan)

Nine statistically significant spatio-temporal clusters were identified ($p < 0.00001$), detailed in Table 1 and mapped in Figure 8. The active temporal window of each cluster is shown in Figure 9.

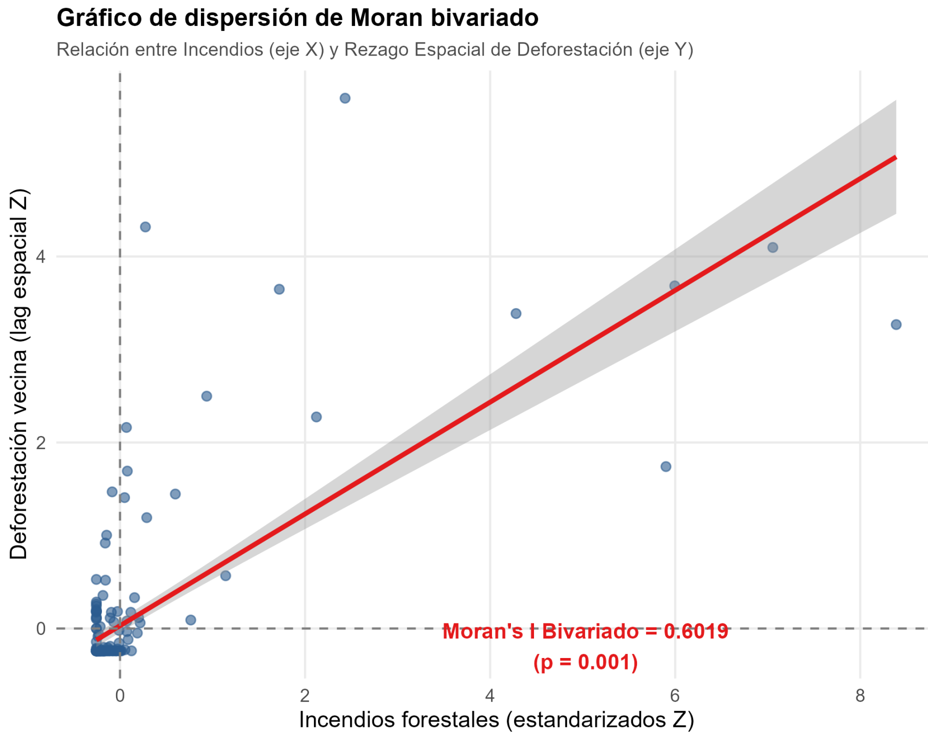

Figure 5: Bivariate Moran scatterplot between fires and the spatial lag of deforestation ($I = 0.6019$, $p = 0.001$).

Table 1: Spatio-temporal forest-fire clusters detected using SaTScan

Cluster	Districts	Period	Radius (km)	Observed	RR	LLR
1	96	2019/1/1 - 2022-12-31	143.9	6273	780.02	35231.2
2	130	2024/1/1 - 2024-12-31	178.1	1059	293.30	4953.5
3	47	2019/1/1 - 2022-12-31	86.9	1263	42.48	3491.5
4	5	2023/1/1 - 2025-12-31	81.5	14264	2.16	2908.3
5	95	2017/1/1 - 2017-12-31	143.5	482	296.55	2261.7
6	4	2018/1/1 - 2021-12-31	95.1	15863	1.80	1933.3
7	2	2019/1/1 - 2022-12-31	31.4	378	11.84	587.5
8	5	2018/1/1 - 2020-12-31	90.6	2064	2.26	521.4
9	21	2017/1/1 - 2017-12-31	54.5	67	184.67	283.0

Cluster 1 (Primary) spans 96 districts at the Cusco–Puno Andean interface (period 2019–2022, radius 143.9 km), standing out for the highest log-likelihood ratio statistic (LLR = 35,231.2) and a **Relative Risk of 780.02** (6,273 observed cases versus 8.8 expected).

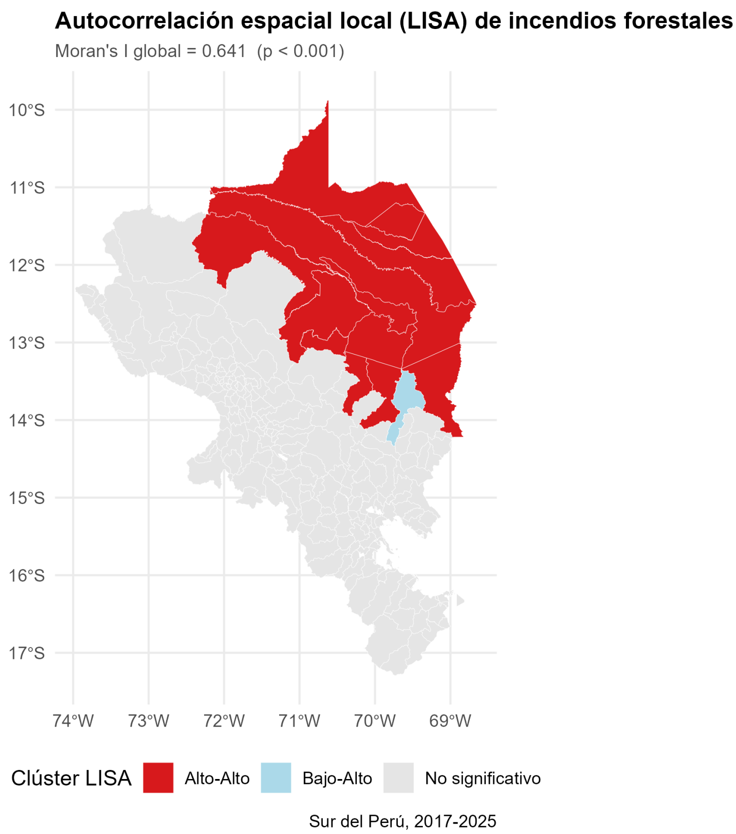

Figure 6: Local spatial autocorrelation map (univariate LISA) of forest fires.

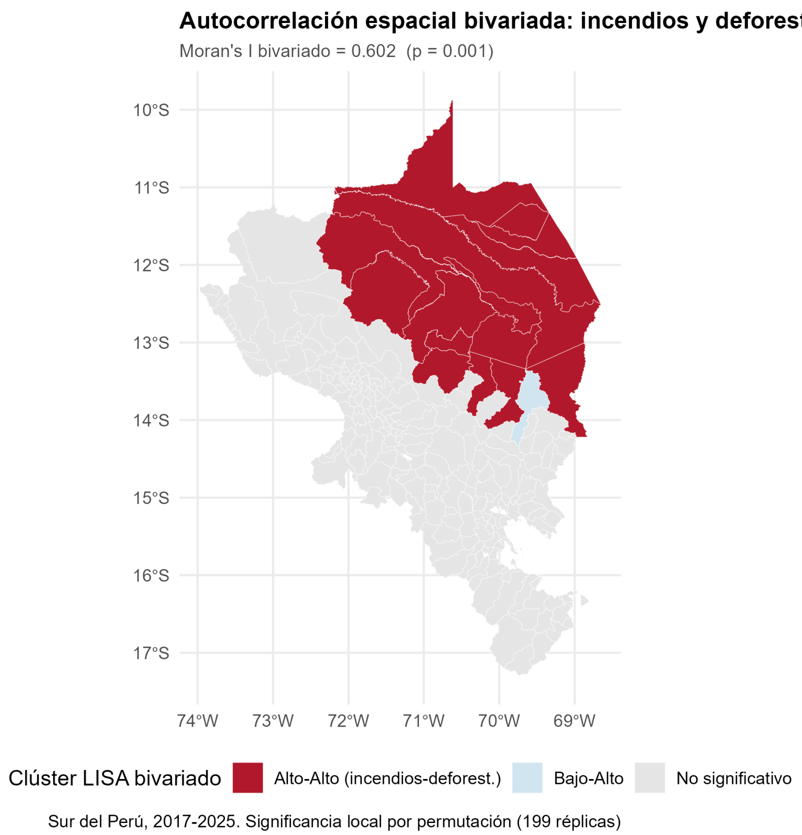

Figure 7: Bivariate LISA map: local association between deforestation and forest fires.

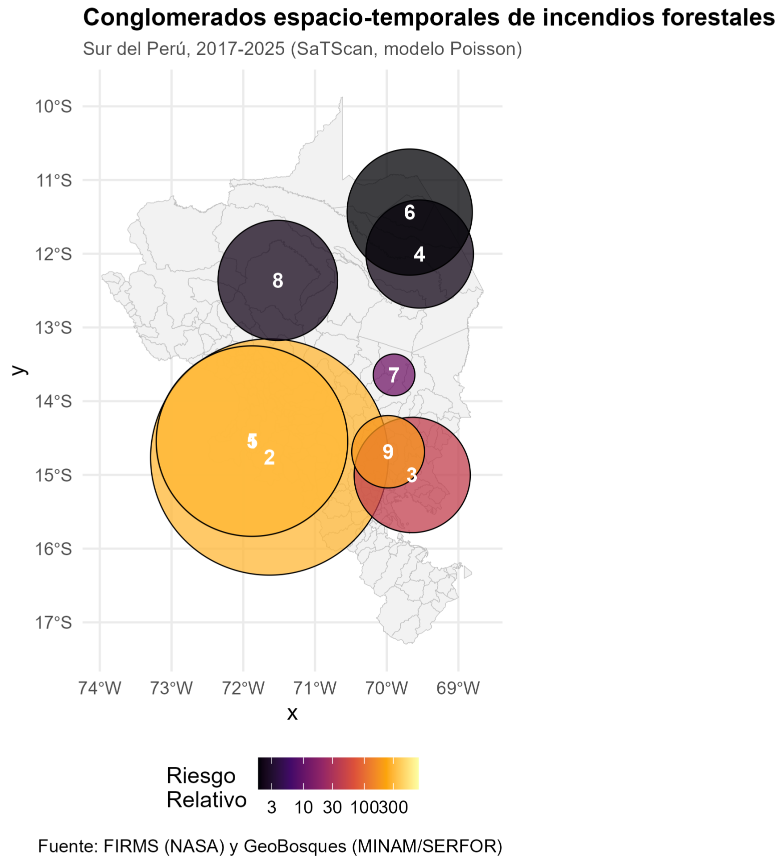

Figure 8: Map of spatio-temporal forest-fire clusters detected by SaTScan (2017–2025).

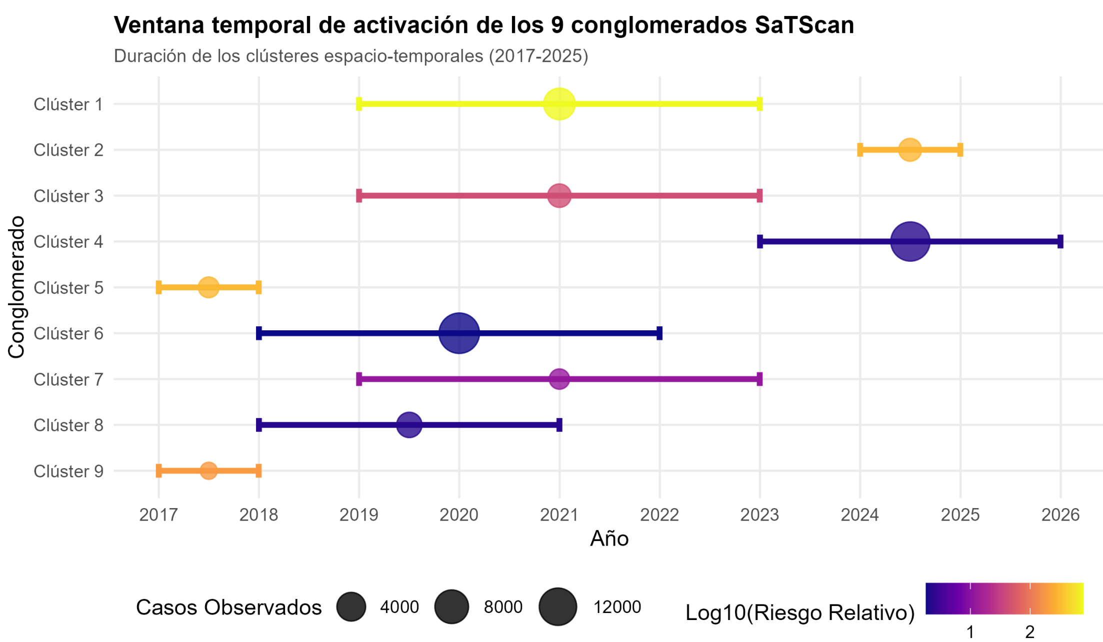

Figure 9: Gantt chart of the active temporal window of the 9 SaTScan clusters.

Cluster 2 groups 130 districts in high-Andean areas of Cusco and Puno during **2024** (RR = 293.30), reflecting the wave of fires associated with that year's extreme drought (Figure 10).

Meanwhile, **Clusters 4 and 6**, in the Amazonian department of Madre de Dios, register the highest absolute burdens: Cluster 4 (5 districts, 2023–2025) totaled **14,264 fires** (RR = 2.16), while Cluster 6 (4 districts, 2018–2021) accumulated **15,863 fires** (RR = 1.80).

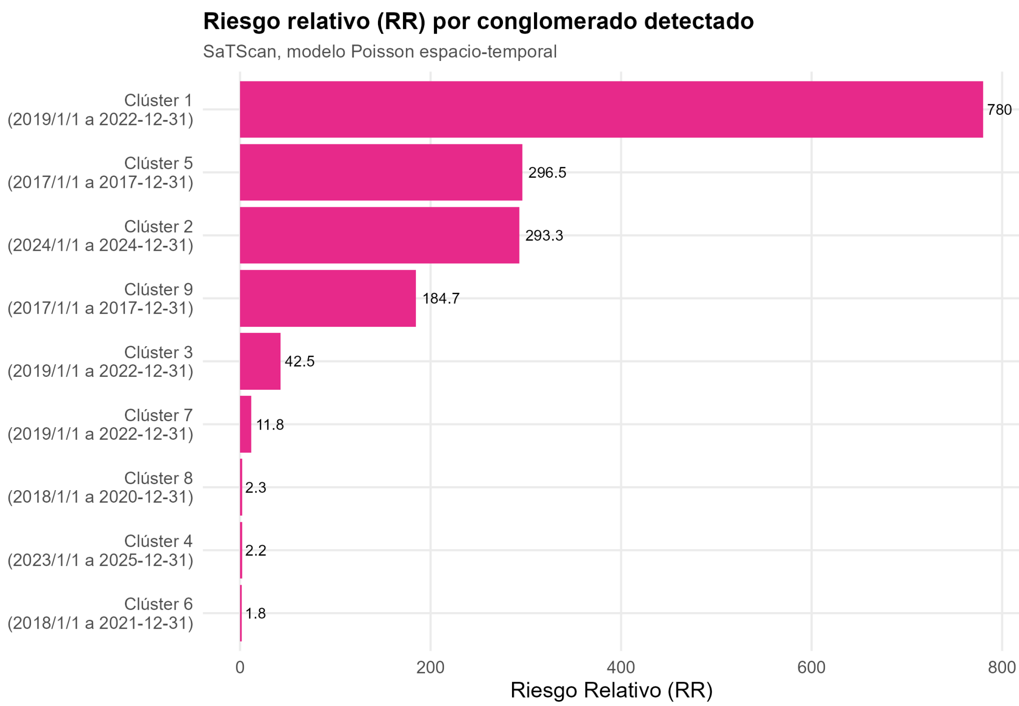

Figure 10: Relative Risk (RR) and significance (LLR) of the 9 spatio-temporal clusters.

4.4. Sensitivity Analysis

Sensitivity tests restricting the maximum spatial window to 25 % and 30 % of the population at risk (Figure 12) demonstrated the stability and robustness of the primary cluster (LLR > 38,000, RR ≈ 772.3), ruling out artifacts from spatial macro-aggregation. The geometry, temporal period, and relative risk of Cluster 1 remained stable across the three window configurations evaluated.

5. Discussion

The findings of this study provide solid empirical evidence of the spatio-temporal heterogeneity of forest fires in southern Peru and are consistent with patterns documented in

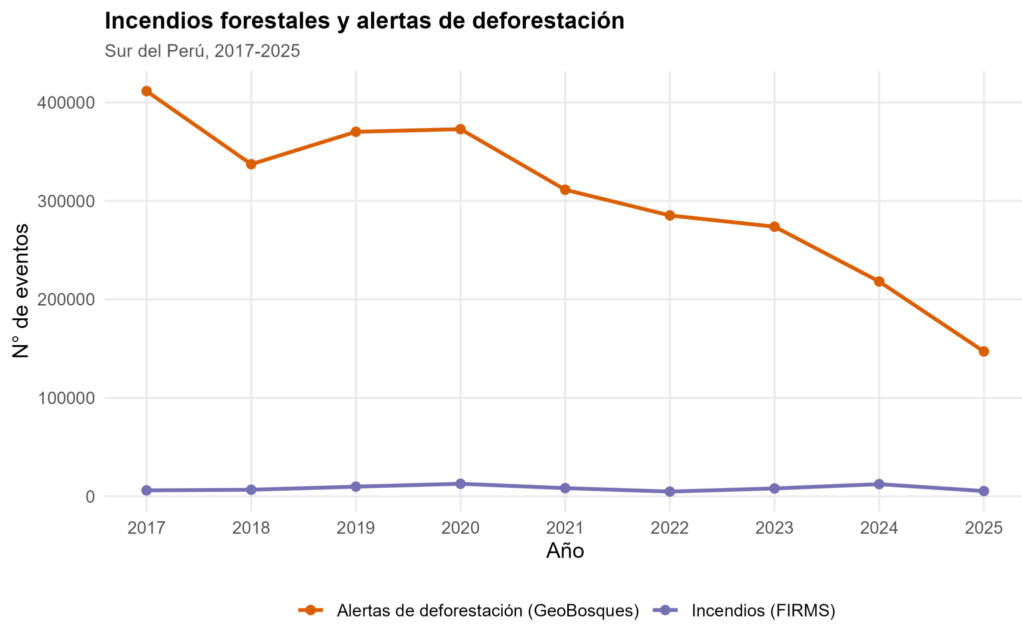

Figure 11: Bivariate relationship between deforestation alerts and forest fires by district.

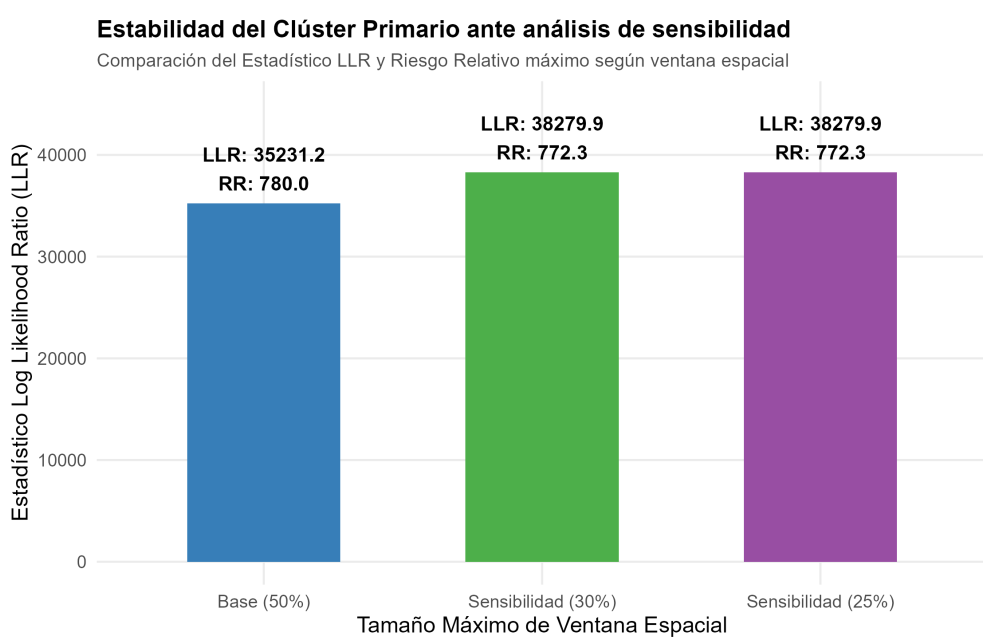

Figure 12: Stability of the Primary Cluster under sensitivity analysis (25 %, 30 %, and 50 % windows).

other ecogeographic contexts of South America. A clear structural duality emerges between two differentiated geographic-ecological patterns:

1. **High-Andean Pattern (Cusco–Puno):** Characterized by clusters with **high Relative Risk (RR from 42 to 780)** but a lower absolute number of hotspots per square kilometer. The burning of natural grasslands and agricultural stubble at the end of the dry season spreads rapidly owing to the strong winds and low relative humidity of the puna. Móstiga et al. (2026) report that the Peruvian Andean region accounts for the largest share of nationally burned area (2.32 %), with a significant increasing trend in fire size ($r = 0.49$, $p = 0.03$).

2. **Amazonian Pattern (Madre de Dios):** Characterized by clusters with **high absolute volume (more than 30,000 combined fires across Clusters 4 and 6)** along the Interoceanic Highway corridor. Massive deforestation from migratory agriculture and mining generates patches of felled biomass that are burned during August and September. This pattern is analogous to that documented by Valente & Laurini (2023) in the Brazilian Amazon, where road proximity is the main spatial driver of fire occurrence.

The strong bivariate autocorrelation ($I = 0.6019$) confirms that land-use change (deforestation) acts as the main spatial catalyst of fire, in line with findings by Ma et al. (2022) for the Amazon, where pasture area and forest-fragmentation edge density are the strongest predictors of fire intensity. Raiol et al. (2026) confirm that pasture expansion significantly increases hotspot rates ($p = 0.04$, Sen's slope = +15.31 hotspots/year). This suggests that fire-prevention policies by SERFOR and MINAM should integrate environmental enforcement against deforestation as a direct measure for reducing forest-fire risk.

Study Limitations

This study presents a set of methodological and data limitations that should be considered when interpreting the results:

(i) MODIS hotspots have a spatial resolution of 1 km, which may result in underdetection of small-extent fires in areas with persistent cloud cover (December–March).

(ii) The VIIRS and MODIS sensors generate independent records for the same thermal event, so there remains a residual risk of hotspot duplication despite the filtering applied.

(iii) The analysis is restricted to 237 districts in southern Peru, limiting generalization at the national level.

(iv) The bivariate Moran's Index measures static spatial association and does not imply causality; spatio-temporal panel models are required to establish formal causal relationships.

(v) The exploratory sensitivity analysis was run with 99 Monte Carlo replications, sufficient for the geometric exploration of clusters but not for reporting definitive p-values for publication.

6. Conclusions

1. **Nine statistically significant spatio-temporal clusters of forest fires** were detected in southern Peru between 2017 and 2025.

2. Fire occurrence exhibits critical seasonality concentrated in the **July–October period (89.51 %)**, peaking in **September (39.01 %)**, with interannual peaks in **2020 and 2024**.

3. Bivariate spatial autocorrelation analysis ($I = 0.6019$, $p = 0.001$) demonstrates a direct spatial association between deforestation alerts and hotspot occurrence in neighboring districts.

4. Sensitivity tests (25 % and 30 %) confirm the robustness of the Cusco–Puno Primary Cluster and of the Amazonian clusters in Madre de Dios under different spatial-window assumptions.

5. These results allow spatial prioritization of Madre de Dios (high absolute event mass) and the Cusco–Puno interface (high relative risk) for the timely deployment of response resources and environmental monitoring by SERFOR and MINAM.

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Data Availability

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